

The Immediate and Retained Effects of One-time Virtual Reality Exposure on Intercultural Sensitivity

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ABSTRACT

This study aims to investigate the immediate and retained effects of one-time virtual reality (VR) exposure on intercultural sensitivity (IS) and identify the contributing factors. Three virtual scenarios about the ethnic minorities in Hong Kong were created for the empirical study. The longitudinal results involving 30 participants (15M 15F) showed that both the immediate and retained effects of the one-time VR exposure on IS were significant. Moreover, linear growth curve models suggested that among the female participants, presence and emotional empathy were closely associated with the change of IS over time, but this relation was not significant among the males.

Index Terms: Human-centered computing—Human computer interaction (HCI)—Interaction paradigms—Virtual reality; Human-centered computing—Human computer interaction (HCI)—Empirical studies in HCI;

1 INTRODUCTION

One of the most widely accepted definitions of intercultural sensitivity (IS) was proposed by Chen and Starosta under the concept of intercultural communication competence, which umbrellaed three dimensions - the cognitive dimension called cultural awareness, the behavioural dimension called intercultural adroitness, and lastly the affective dimension called intercultural sensitivity [5]. Given the potential of VR in enhancing emotional empathy [13] and the connection between emotional empathy and IS [2, 3, 6], there is a recent interest in using VR for enhancing IS (e.g., [1, 7, 12]). However, prior studies often focused on the immediate effects right after the VR exposure but failed to investigate whether the effects could be retained. Also, in these studies, the effects are often attributed to long-term exposure to the media or intercultural environments; there is a lack of studies that focus on the effects of one-time VR exposure on IS. Moreover, the factors contributing to the effects of VR exposure on IS are underexplored. The present study aims to address these literature gaps.

2 RELATED WORK

2.1 Intercultural Sensitivity

The definition of IS is constantly evolving. Bennett adopted a developmental approach to describe IS as a process of transforming oneself from ethnocentric stages to ethnorelative stages [3]. The change involves the affective, cognitive, and behavioural aspects of intercultural interaction. Chen and Starosta placed a greater emphasis on the affective side. They defined IS as the affective component in intercultural communication competence, a mindset that allows individuals to recognize the behavioural, perceptual, and emotional

differences in their counterparts from different cultures [6]. As IS is crucial to effective intercultural communication, researchers have investigated the effectiveness of various training approaches to enhance IS, such as by movies [11] and by creative imagination [9]. However, there are limited studies examining the effectiveness of training in enhancing IS using immersive media.

2.2 Presence

Presence is often referred to as *the sense of being there*, although it could be interpreted slightly differently in different contexts [15]. Presence is now widely regarded as the desired design outcome of VR experiences thus a metric for measuring the quality of the experiences; it is also considered a perceptual illusion that is quite unique to VR exposure [15]. Although it is unclear whether presence can predict the effects of VR exposure on IS, its relationship with empathy, especially emotional empathy, has been studied. Prior studies suggest that a high level of presence experienced by participants can be related to increased emotional reactivities [10, 16]. Bertrand et al. found that such emotional reactivities, which attracted their participants' full attention on the immersive and interactive storytelling, could improve the participants' empathy [4].

3 METHOD

3.1 Experimental Design and Procedure

The study employed the repeated measures design across three time points, T_1 , T_2 , and T_3 . Specifically, T_1 is right before the VR exposure, T_2 is right after the exposure, and T_3 is four weeks after the exposure. The participants' IS, which is the dependent variable of this study, was measured at all three time points. Emotional empathy, which is dispositional, was measured at T_1 . Presence was measured at T_2 .

3.2 Participants

A total of 34 individuals registered for participation through the recruitment flyers posted on campus. Two participants withdrew from the experiment due to time constraints. Two participants were excluded from the data analyses as they did not participate in the tests at T_3 . Among the 30 participants included for analyses, the female-male ratio is 1:1, and they were aged between 18 and 30 ($Mean = 22.56$, $SD = 3.57$). All of them had normal or correct-to-normal vision. None of them reported health conditions that might prevent them from participation.

3.3 Materials and Apparatus

The materials used for this study are the three virtual scenarios created based upon several typical locations reflecting the daily lives of the ethnic minorities in Hong Kong, which were adapted from [12]. Using gamification techniques, the discoverable spots and items are embedded in the scenarios like a classic puzzle game. Players need to navigate in the scenarios using the classical teleportation method and once they are close to the spots and items, visual cues will light up and aid them to interact with the discoverables and learn the conveyed information. The scenarios were delivered through an Oculus Rift S head-mounted display (HMD) in the standing mode.

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Table 1: Descriptive statistics of the IRI-EC, ISS, and IPQ scores.

	T_1	T_2	T_3
	Mean (SD)	Mean (SD)	Mean (SD)
IRI-EC			
Male	17.13 (3.93)		
Female	19.40 (4.14)		
Combined	18.27 (4.13)		
ISS			
Male	3.98 (0.33)	4.01 (0.42)	4.00 (0.34)
Female	3.68 (0.26)	3.87 (0.28)	3.86 (0.31)
Combined	3.83 (0.33)	3.94 (0.36)	3.93 (0.32)
IPQ			
Male		3.81 (0.59)	
Female		3.57 (0.65)	
Combined		3.69 (0.62)	

3.4 Measures

The Intercultural Sensitivity Scale (ISS) [6] was adopted to measure the participants' IS. The current-sample reliability is .772 (Cronbach's α) at T_1 , .841 at T_2 , and .837 at T_3 . The Empathic Concern subscale of the Interpersonal Reactivity Index (IRI-EC) [8] was used to measure the participants' emotional empathy. Other subscales were not administered as our theoretical framework focuses on emotional empathy only. The reliability is .661 (Cronbach's α) at T_1 . The Igroup Presence Questionnaire (IPQ) [14] was used to assess the sense of presence. The reliability is .701 (Cronbach's α) at T_2 .

3.5 Data Analysis

To test the effect of one-time VR exposure on ISS, the ISS scores measured at three time points were calculated to be the dependable variables for repeated measures ANOVA with gender as the between-subject factor in IBM SPSS Statistics 26. Latent growth curve models were also conducted to inspect the effect of emotional empathy and presence on the change in IS. The models were fitted using Mplus version 8.

4 RESULTS

Table 1 shows the descriptive statistics of the results. A mixed repeated measures ANOVA with time as within-subject factor and gender as between-subject factor were performed to test the effect of time and interaction effect of time $\times$ gender on IS. The analysis revealed that the main effect of time was significant, $F(2,56) = 5.75$, $p = .005$, $\eta_p^2 = .170$. Post-hoc analysis with a Bonferroni adjustment revealed that IS significantly increased from T_1 to T_2 , $p = .029$, 95% CI $[-.21, -.10]$, and there was no significant difference between T_2 and T_3 , $p = 1.00$, 95% CI $[-.07, .08]$, but the increase from T_1 to T_3 was significant $p = .046$, 95% CI $[-.21, -.002]$.

Among the female participants, the linear growth curve model with emotional empathy as time-invariant covariate and presence as time-varying covariate regressed on T_2 and T_3 ISS scores had a good model fit ($\chi^2(3) = .531$, $p = .912$; $CFI = 1.00$; $TLI = 1.225$; $RMSEA < .001$, 90% CI $[-.000, .167]$). The effect of emotional empathy on the intercept ($B = .016$, $SE = .016$, $p = .309$) and slope ($B = -.007$, $SE = .008$, $p = .347$) was not statistically significant. As for presence, it could significantly predict IS at T_2 ($B = .148$, $SE = .047$, $p = .002$) T_3 ($B = .244$, $SE = .092$, $p = .008$). Among the male participants, the same model fitted poorly.

5 DISCUSSION AND CONCLUSION

Following the definition of IS and its close relation with emotional empathy, we designed this longitudinal study to investigate the effects of one-time VR exposure on IS. It was found that the effects were significant and could be retained for at least four weeks after the

exposure. The linear growth curve models showed that among the female participants, the change of IS over three time points was closely associated with presence and emotional empathy. However, the relationship was not significant among the male participants. These findings provided valuable insights and practical implications regarding the future design of VR experiences for IS training. Specifically, the close association among IS, presence, and emotional empathy suggests that designing VR experiences for IS training can directly benefit from the prior studies on how to better enhance the sense of presence during VR exposure (e.g., [14]). Meanwhile, our study is mainly limited in the reliability of some measures (e.g., IRI-EC) among the relatively small number of participants, due to the difficulty of participant recruitment during the COVID-19 pandemic. We suggest future research should consider the gender effects when designing empirical studies and further explore the gender effects on larger data sets.

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